MACHINE LEARNING · PROJECT REVIEW 3 · UNIT 5

Ensembles

EarlyDrop, PBL-III. Combining models to flag academic risk: trees, bagging, boosting, Bayesian model averaging and stacking.

● SYNTHETIC COHORT · n = 400 · SAME FOLDS AS PBL-I

Faculty · Dr. Santhiya Parivallal

Yash Arun Mulajkar
RA2411003011562

Bignesh Sahu
RA2411003011563

Shaurya Ojha
RA2411003011565

Aditya Bhutada
RA2411003011607

PROBLEM IDENTIFICATION

Can a committee of models beat the best single one?

TARGET (UNCHANGED)

Flag students whose **end-semester SGPA will fall below 6.5**, from mid-semester survey answers.

27.8%

of students are at risk

BAR TO BEAT · PBL-I

Logistic regression, stratified 5-fold CV.

0.953

CV ROC-AUC, the best of seven Unit 2 classifiers

UNIT 5 QUESTIONS

- Does averaging or boosting weak learners lift accuracy?
- Is any gain **statistically** real?
- Why do ensembles help, when they do?

DATASET SELECTION & EVALUATION PROTOCOL

Same students, same features, same folds.

COHORT

400

synthetic students in the live form's 50-column schema, calibrated on 15 real responses

FEATURES

15

grades, attendance, backlogs, study habits, motivation, stress

HELD-OUT TEST

100

stratified 25 % split for curves, ROC and bias–variance (train 300)

MODELS COMPARED

17

5 single models and 12 ensembles, 6 written from scratch

PROTOCOL

Stratified 5-fold CV with seed 42 (the **same folds as Unit 2**) · out-of-fold ROC-AUC, accuracy, recall, Brier, log-loss · significance by repeated 5 × 3 CV with the **Nadeau–Bengio corrected t-test** · every from-scratch model is checked against scikit-learn.

UNIT 5 · COMBINING MODELS

PBL-III

Six ways to combine

CART · Gini · pruning

Bagging · random forest · OOB

AdaBoost · exponential loss

Gradient boosting · shrinkage

Bayesian model averaging

Generalized additive models

Adaptive basis functions · MLP

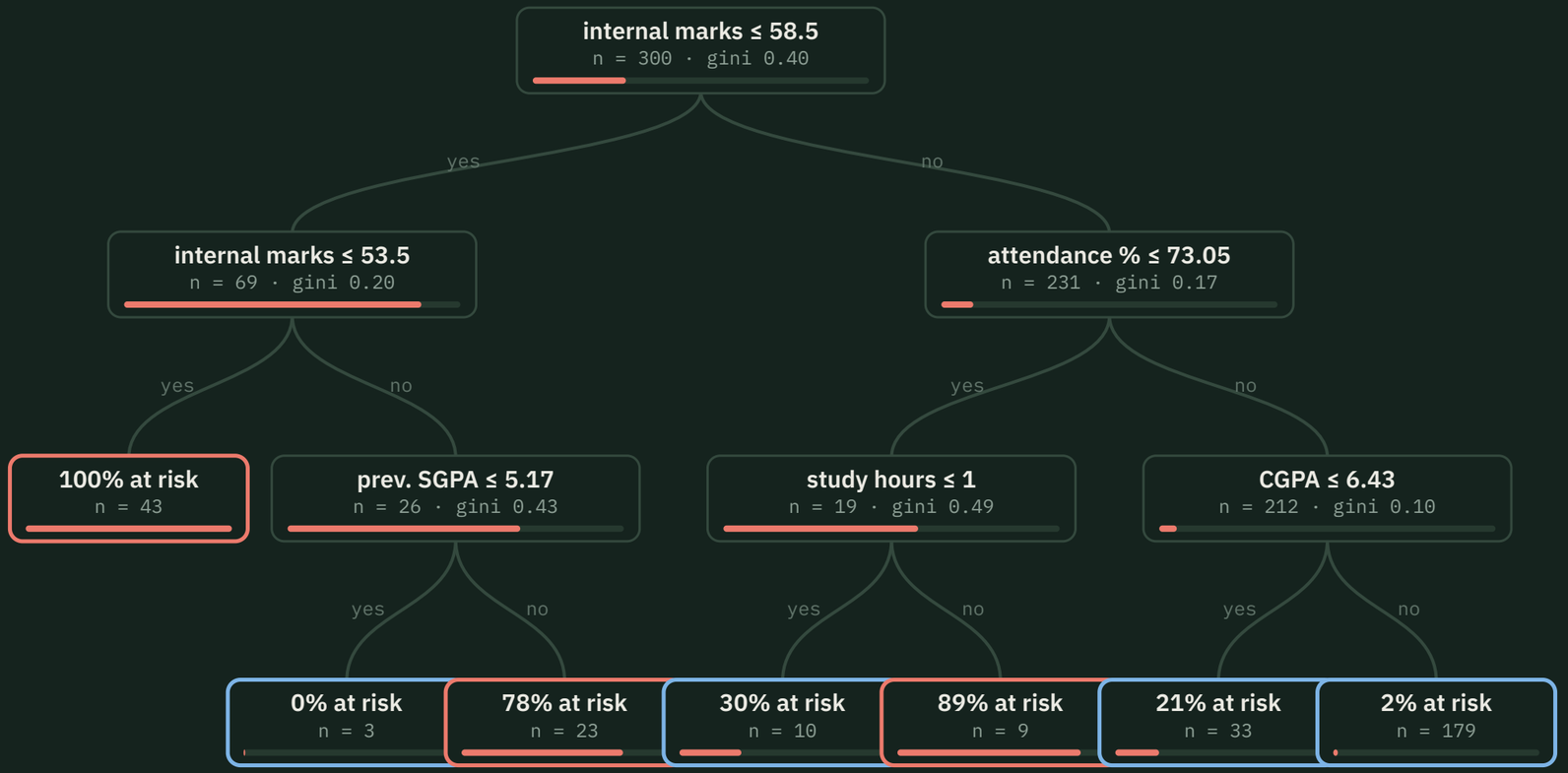
Voting · stacking

From scratch: CART, bagging, random forest, AdaBoost, gradient boosting, BMA. Library: extra trees, histogram GBM, GAM, MLP, voting, stacking.

UNIT 5 · CLASSIFICATION AND REGRESSION TREES

A tree asks one question at a time.

SCRATCH CART ON THE TRAINING SET · DEPTH 3 · LEAF BORDER CORAL = PREDICT AT RISK



$$\text{Gini}(t) = 1 - \Sigma_k\, p_k^2$$

$$\Delta = G(t) - (n_L/n)G(L) - (n_R/n)G(R)$$

GREEDY, EXHAUSTIVE

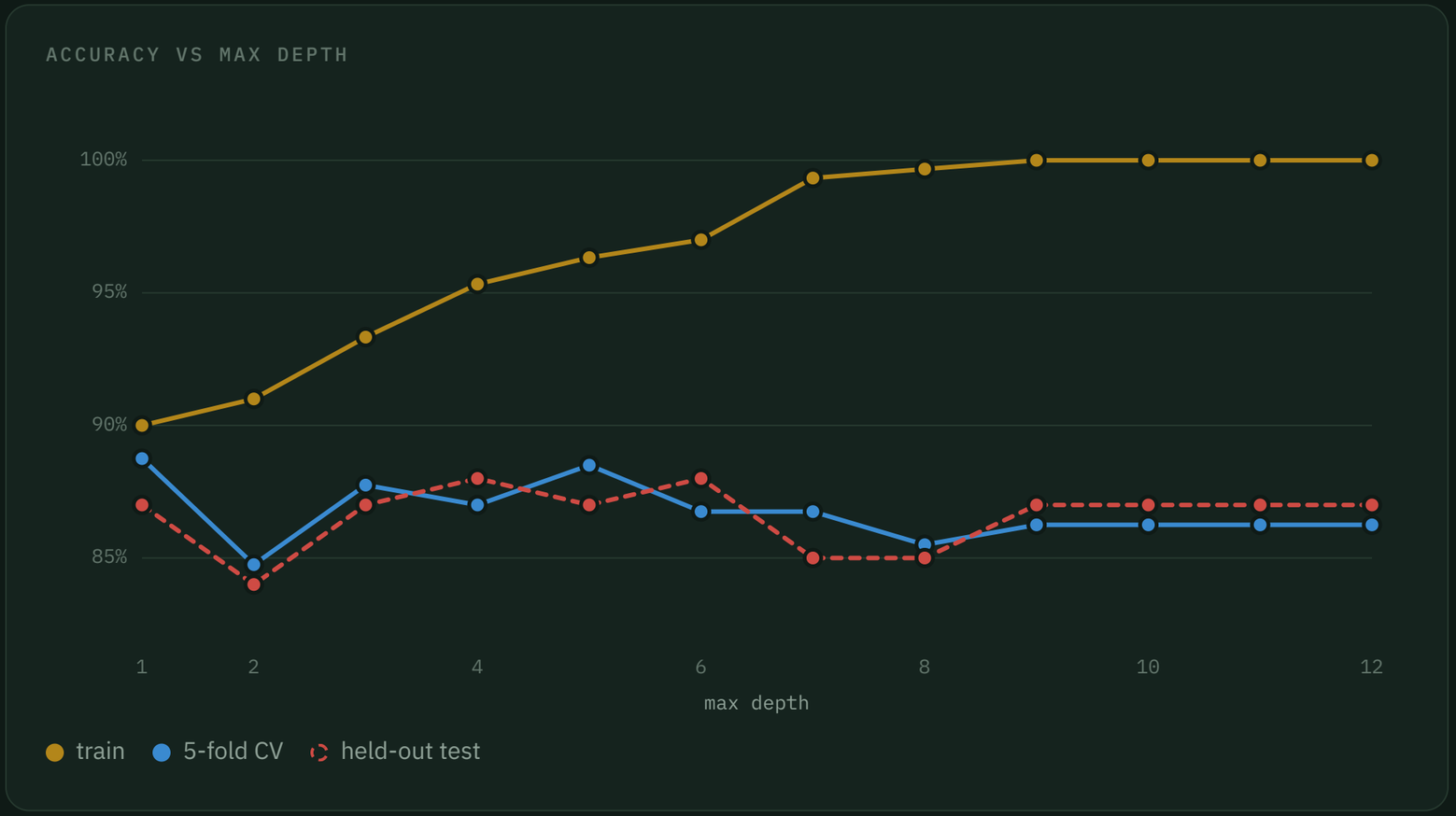
Every feature, every mid-point threshold, keep the split with the largest Δ. Recurse until leaves are pure.

CHECK VS SKLEARN (DEPTH 4)

Same root split · **100%** identical test predictions

UNIT 5 · OVER-FITTING AND COST-COMPLEXITY PRUNING

One tree memorises. That is the problem ensembles solve.



$$R_\alpha(T) = R(T) + \alpha \cdot |\text{leaves}(T)|$$

PRUNING (α BY 5-FOLD CV)

28 → 3 leaves

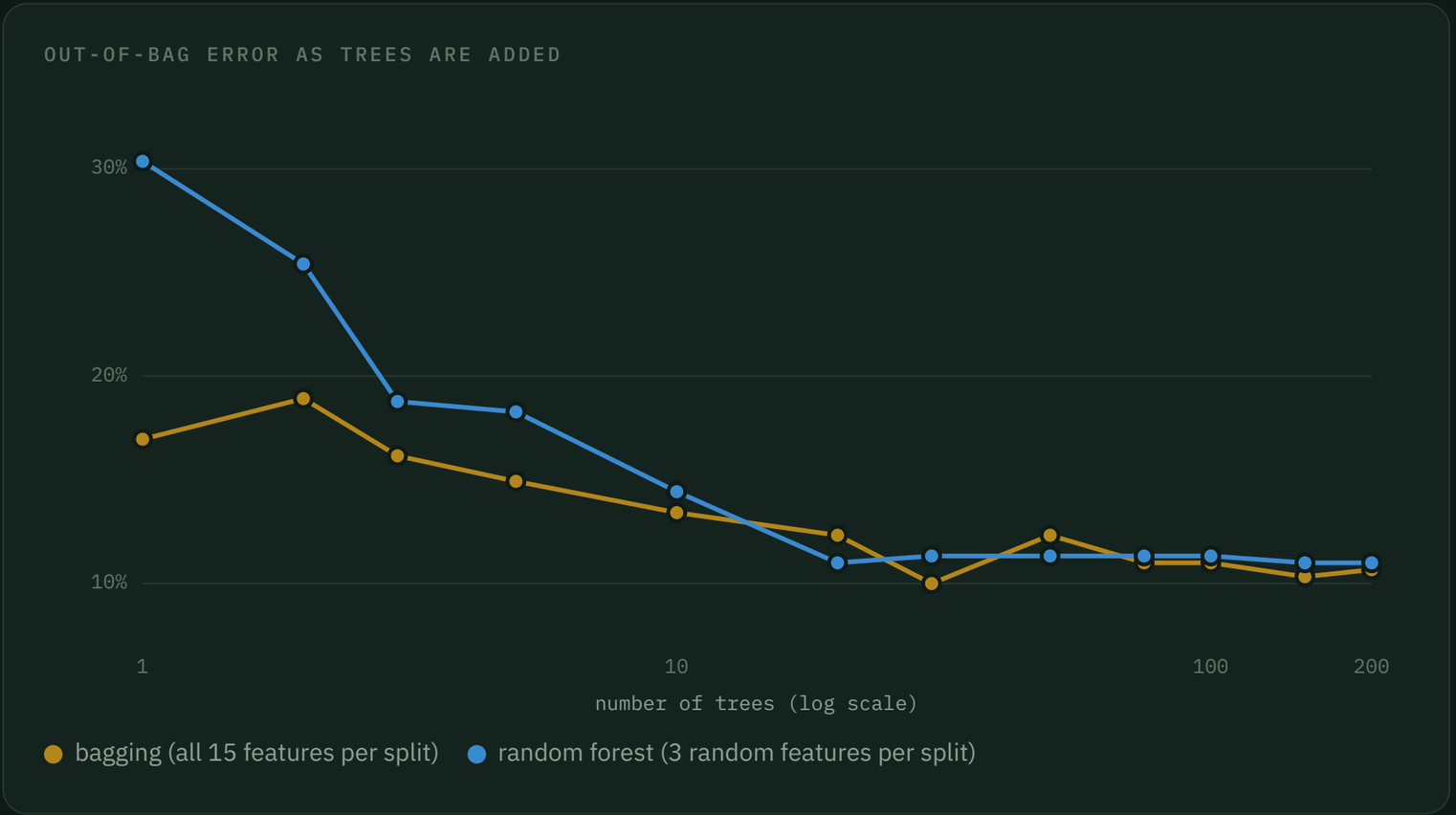
α = 0.0125

5-FOLD CV ROC-AUC

full tree **0.821** · pruned **0.863** · logistic **0.953**

UNIT 5 · BAGGING AND RANDOM FORESTS (SCRATCH)

Average many noisy trees and the noise cancels.



$$\hat{f}_{bag}(x) = (1/B)\ \Sigma_b\ \hat{f}_b(x)$$
$$\text{Var} = \rho\sigma^2 + (1 - \rho)\sigma^2/B$$

TREE-TO-TREE CORRELATION ρ

0.70 bagging

0.61 random forest (√15 features)

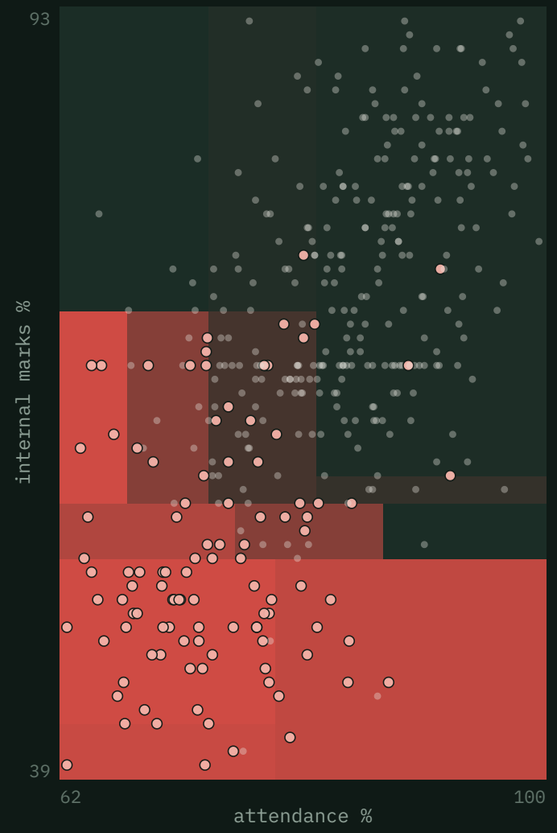
OOB ACCURACY

scratch RF **89.0%** · sklearn RF **89.3%**

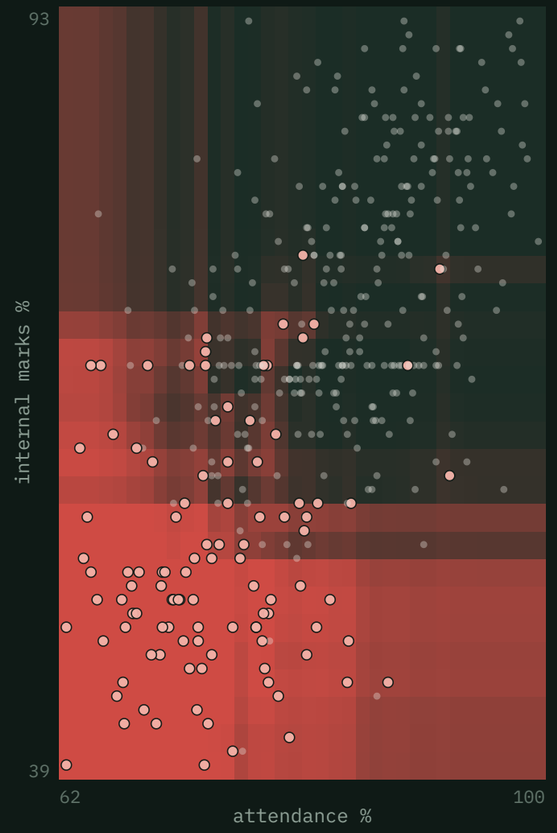
UNIT 5 · WHAT THE MODELS LEARN

Staircase vs smooth: two features, four learners.

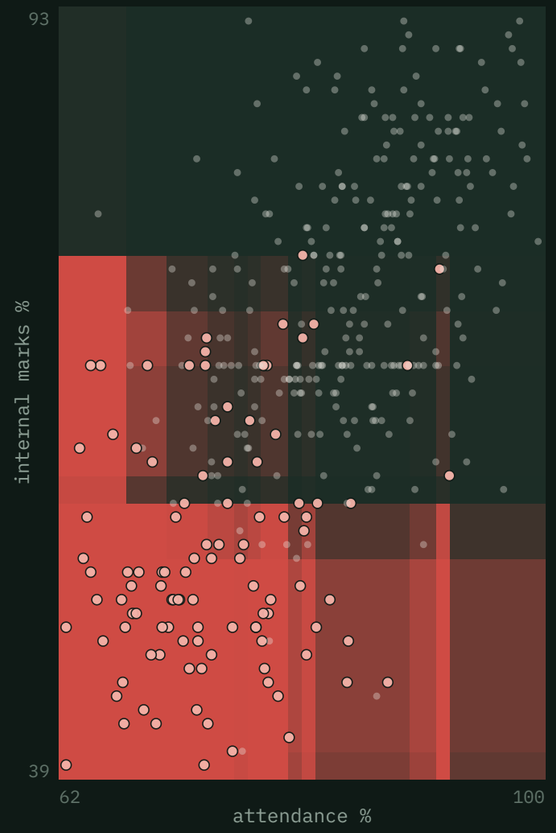
CART (depth 4) · train 91%



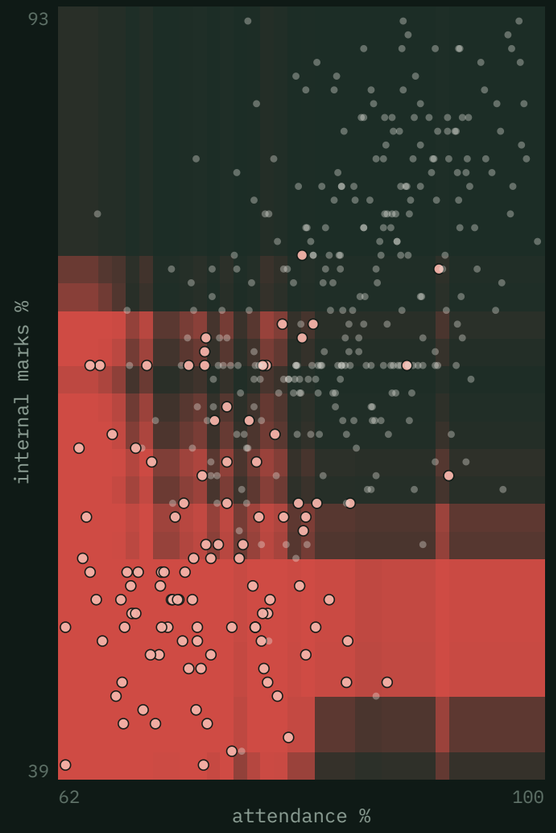
Random Forest · train 95%



AdaBoost · train 91%



Gradient Boosting · train 94%

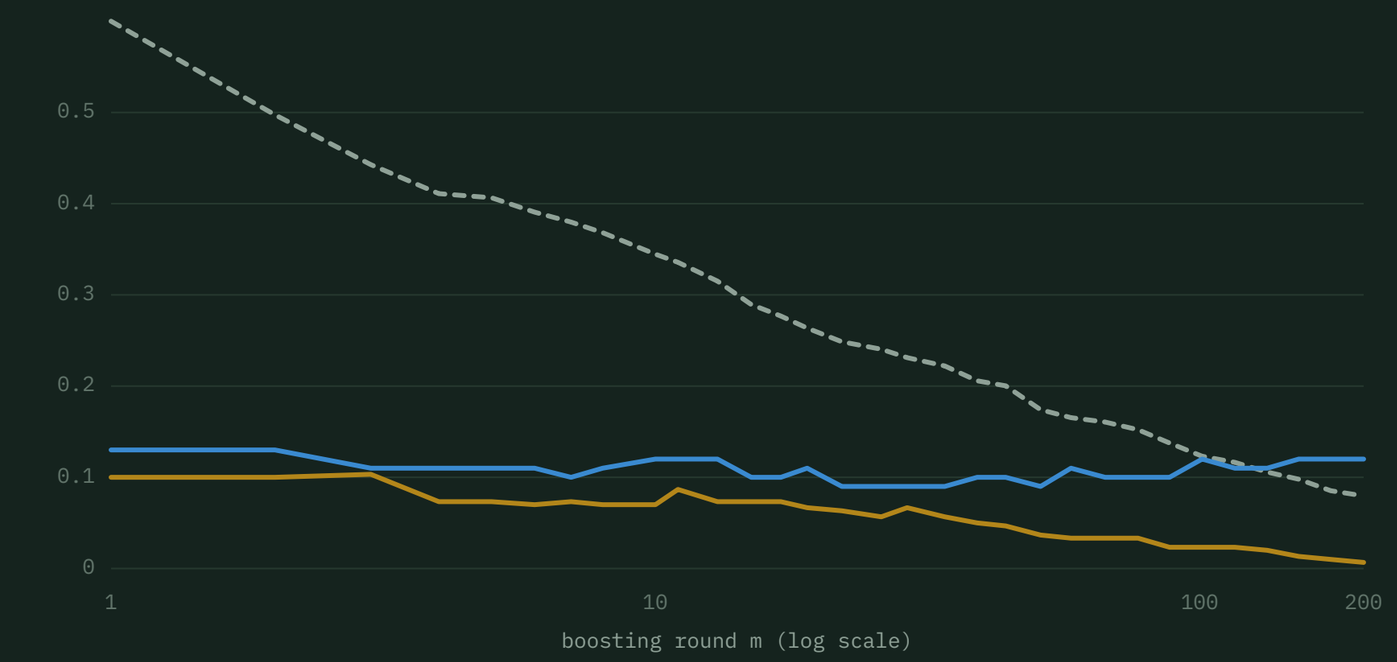


Colour = predicted P(at risk) from dark (0) to red (1) · bright dots = students actually at risk · trained on attendance and internal marks only.

UNIT 5 · BOOSTING (ADABOOST FROM SCRATCH)

Each new stump focuses on the students the last ones missed.

ERROR VS BOOSTING ROUNDS · 200 DECISION STUMPS



$$\varepsilon_m = \Sigma\, w_n\, I(y_m(x_n) \neq t_n) \;/\; \Sigma\, w_n$$
$$\alpha_m = \ln((1 - \varepsilon_m)/\varepsilon_m)$$
$$w_n \leftarrow w_n \exp(\alpha_m\, I(\text{miss}))$$

FIRST STUMP

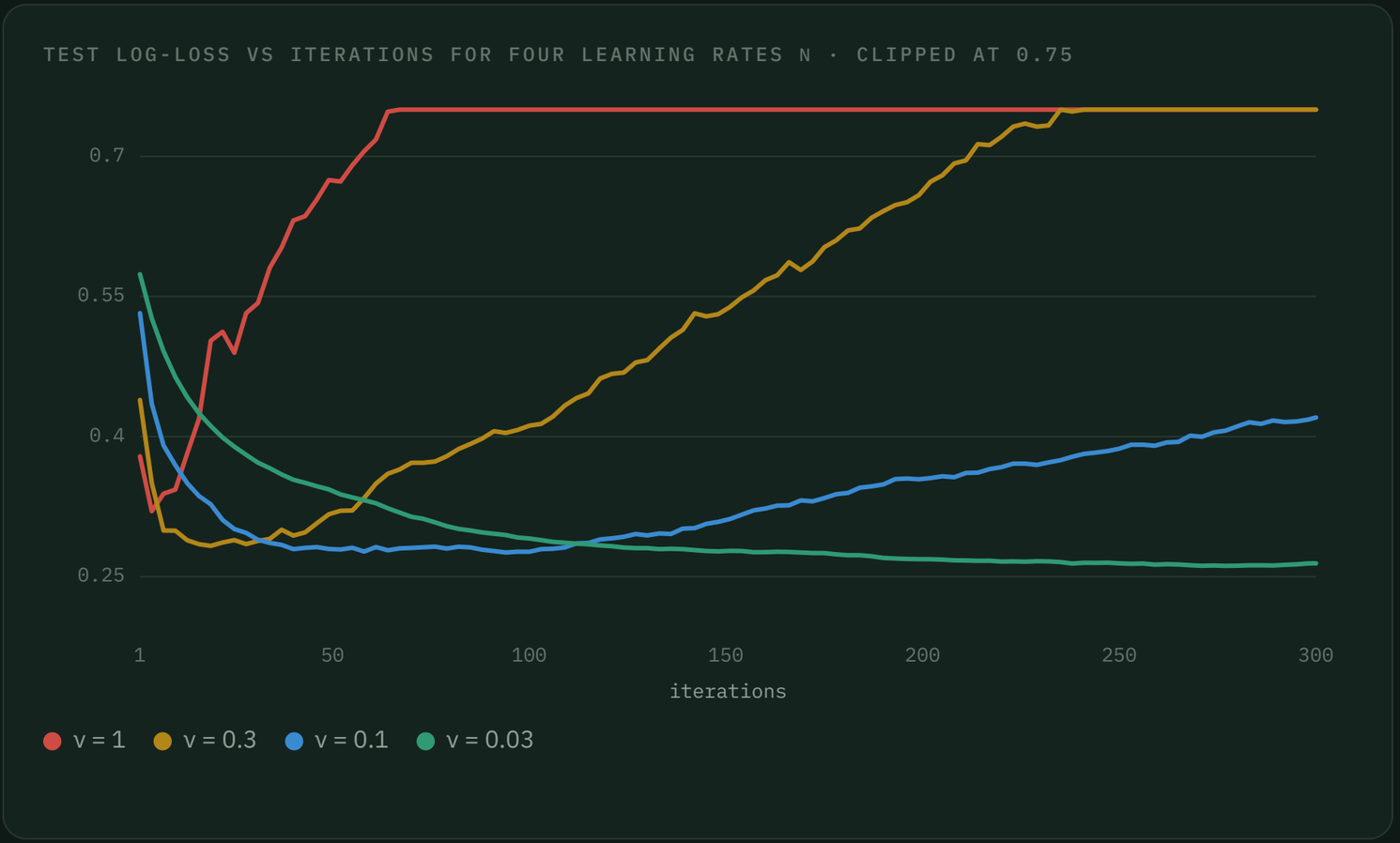
internal marks ≤ 58.5 · ε = 0.100 · α = 2.20

CHECK VS SKLEARN (SAMME)

100% same test predictions · decision correlation 1.000

UNIT 5 · GRADIENT BOOSTING (SCRATCH)

Small steps, more trees, less over-fitting.



$$r_n = y_n - \sigma(F(x_n))$$
$$\gamma_{leaf} = \Sigma\ r\ /\ \Sigma\ p(1-p)$$
$$F \leftarrow F + \nu \cdot \gamma_{leaf}(x)$$

BEST ITERATION

N	BEST M	TEST LOG-LOSS
1	8	0.313
0.3	23	0.281
0.1	98	0.275
0.03	279	0.261

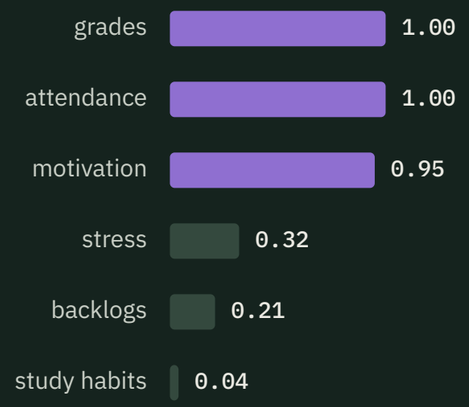
CHECK VS SKLEARN

max |Δp| = **0.0035** · AUC 0.946 vs 0.946

UNIT 5 · BAYESIAN MODEL AVERAGING (SCRATCH)

Average over 64 models, weighted by evidence.

POSTERIOR INCLUSION PROBABILITY



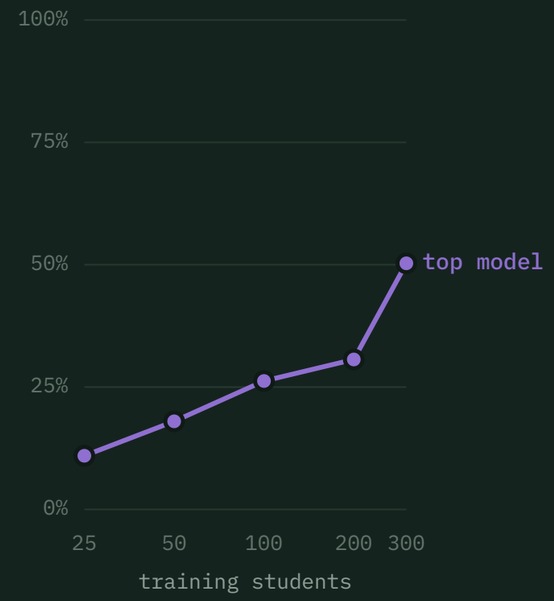
$$p(y|x,D) = \Sigma_M \, p(y|x,M,D) \, p(M|D)$$

MOST PROBABLE MODELS · LAPLACE EVIDENCE

FEATURE BLOCKS IN THE MODEL	P(M | D)
grades + attendance + motivation	50.3%
grades + attendance + motivation + stress	22.5%
grades + attendance + backlogs + motivation	13.2%
grades + attendance + backlogs + motivation + stress	6.2%
grades + attendance + stress	1.6%

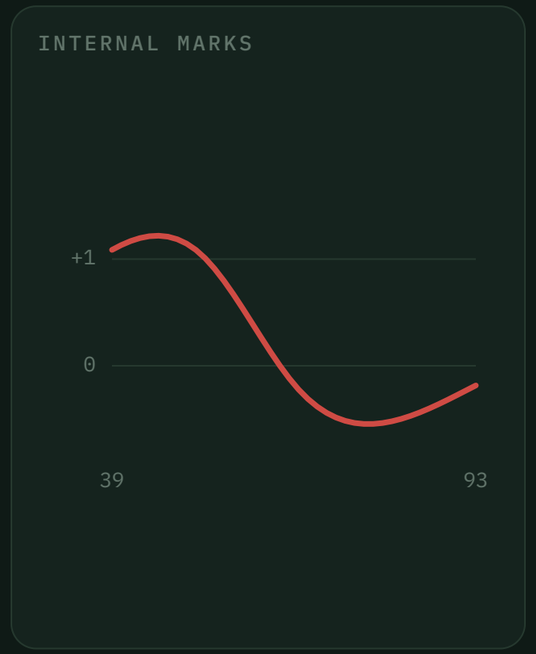
Effective number of models **3.1** · the full model ranks 14 of 64

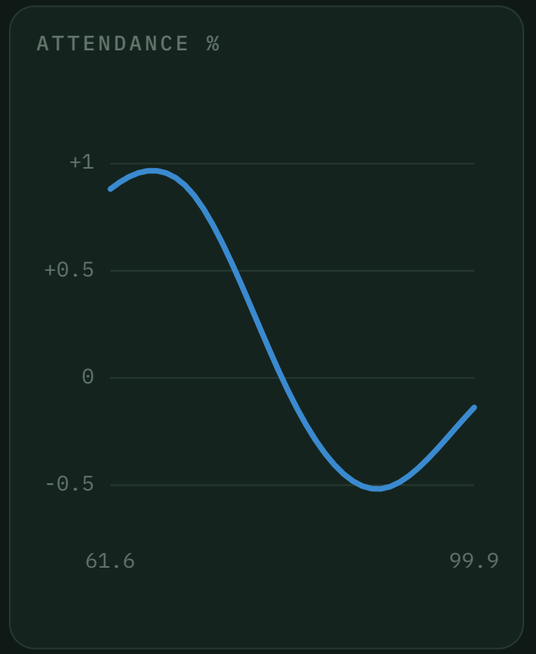
THE POSTERIOR CONCENTRATES AS N GROWS

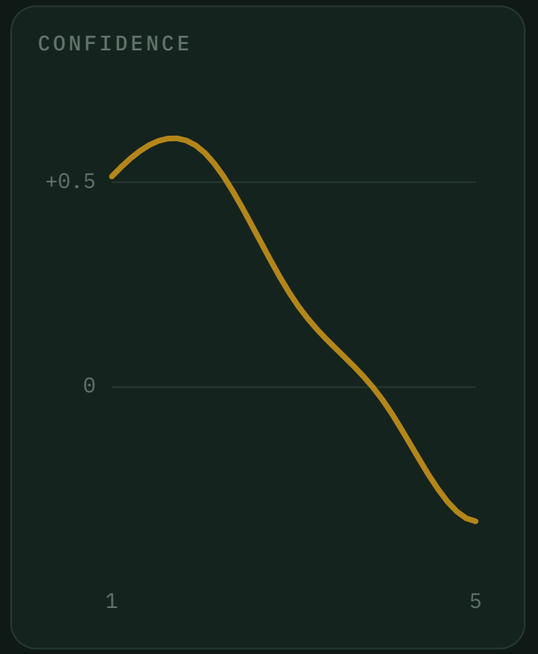


UNIT 5 · GENERALIZED ADDITIVE MODELS

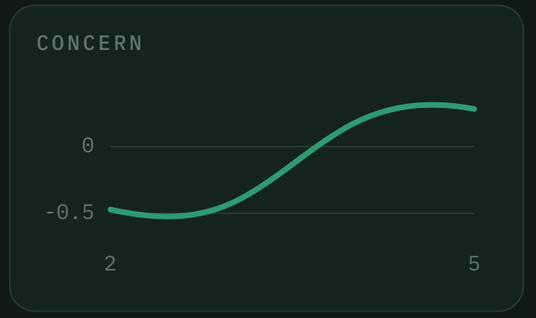
One smooth curve per feature, still readable.













$$\text{logit } p(x) = \beta_0 + \Sigma_j f_j(x_j)$$

BASIS

cubic B-splines, 5 knots → 7 basis functions per feature, L2 penalty C = 0.3

HELD-OUT ROC-AUC

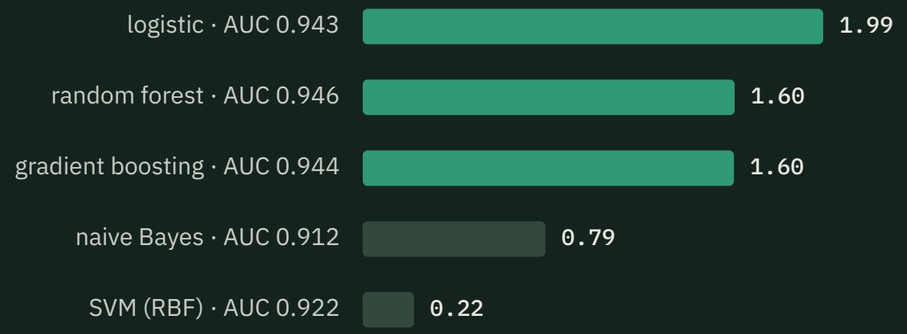
0.967

partial log-odds, centred on the training mean

UNIT 5 · VOTING AND STACKING

Stacking learns the weights. Correlated models leave little to learn.

META-LEARNER WEIGHTS (LOGISTIC ON BASE PROBABILITIES)



Soft voting averages the same probabilities with equal weights.

CORRELATION OF OUT-OF-FOLD BASE PREDICTIONS

	LR	SVM	NB	RF	GB
LR	1.00	0.95	0.81	0.94	0.93
SVM	0.95	1.00	0.86	0.95	0.92
NB	0.81	0.86	1.00	0.88	0.82
RF	0.94	0.95	0.88	1.00	0.96
GB	0.93	0.92	0.82	0.96	1.00

IMPLEMENTATION · VERIFICATION

Written from scratch, checked against scikit-learn.

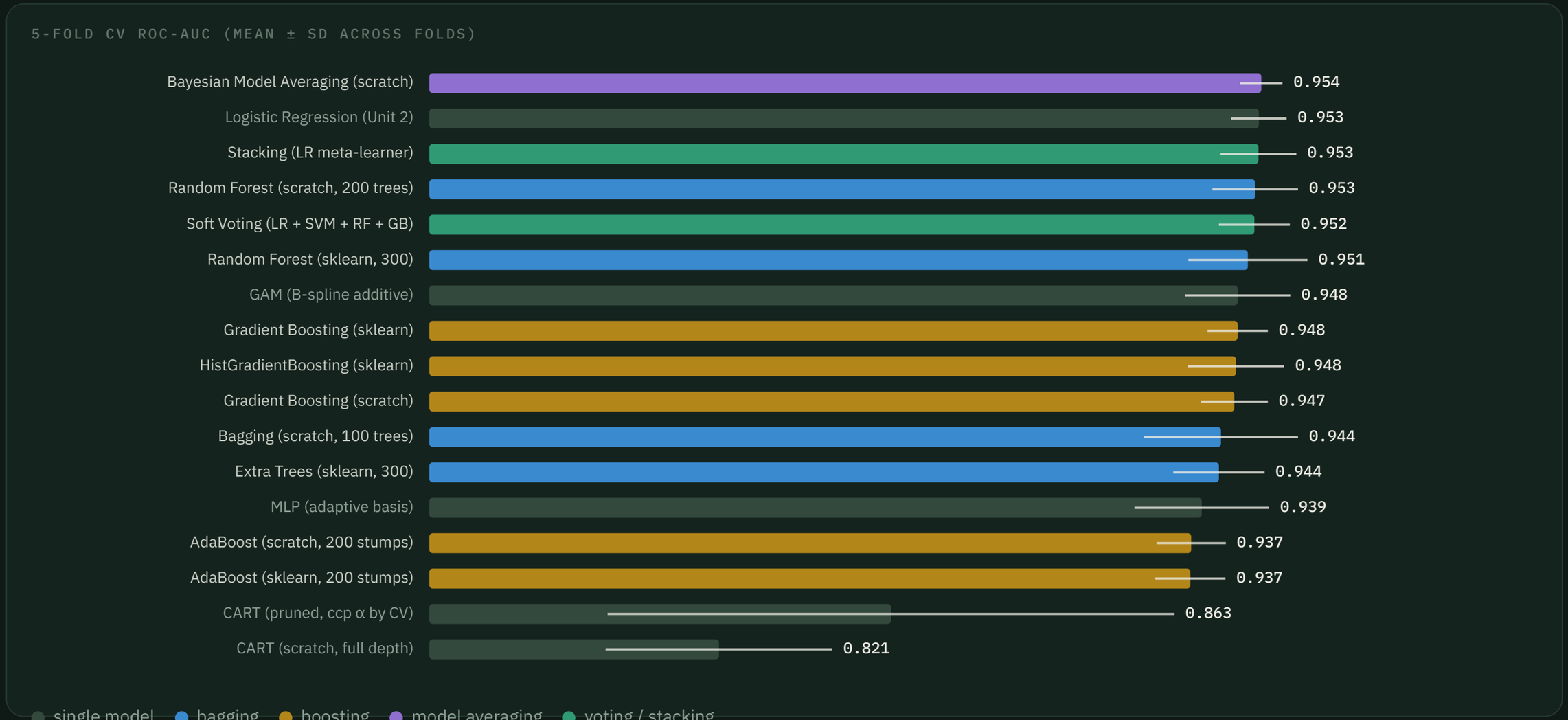
SCRATCH MODEL	CHECKED AGAINST	EVIDENCE	RESULT
CART, depth 4	DecisionTreeClassifier	same root split: internal marks ≤ 58.5	100% same test predictions
CART, full depth	DecisionTreeClassifier	27 vs 28 leaves · depth 9 vs 9	95% agreement (ties break differently)
Random forest, 200 trees	RandomForestClassifier	OOB accuracy 89.0% vs 89.3%	test AUC 0.954 vs 0.959
AdaBoost, 200 stumps	AdaBoostClassifier (SAMME)	decision-function correlation 1.000	100% same predictions
Gradient boosting, 150 trees	GradientBoostingClassifier	max |Δp| = 0.0035	test AUC 0.946 = 0.946
Bayesian model averaging	Laplace evidence (Bishop 4.137)	64 Bayesian logistic models	test AUC 0.959

PIPELINE SELF-CHECKS

- ✓ schema 49 modelling columns
- ✓ no missing after cleaning
- ✓ ols normal equations match sklearn
- ✓ bayesian lr auc close to sklearn
- ✓ scratch gmm matches sklearn ari≥ 0.95
- ✓ scratch pca eigenvalues match sklearn
- ✓ scratch hmm loglik matches hmmlearn
- ✓ scratch hmm as good as hmmlearn fit
- ✓ scratch cart matches sklearn tree
- ✓ scratch adaboost matches sklearn
- ✓ scratch gboost matches sklearn
- ✓ scratch random forest auc close to sklearn

COMPARATIVE PERFORMANCE ANALYSIS

Seventeen models, same five folds.



COMPARATIVE ANALYSIS · SIGNIFICANCE

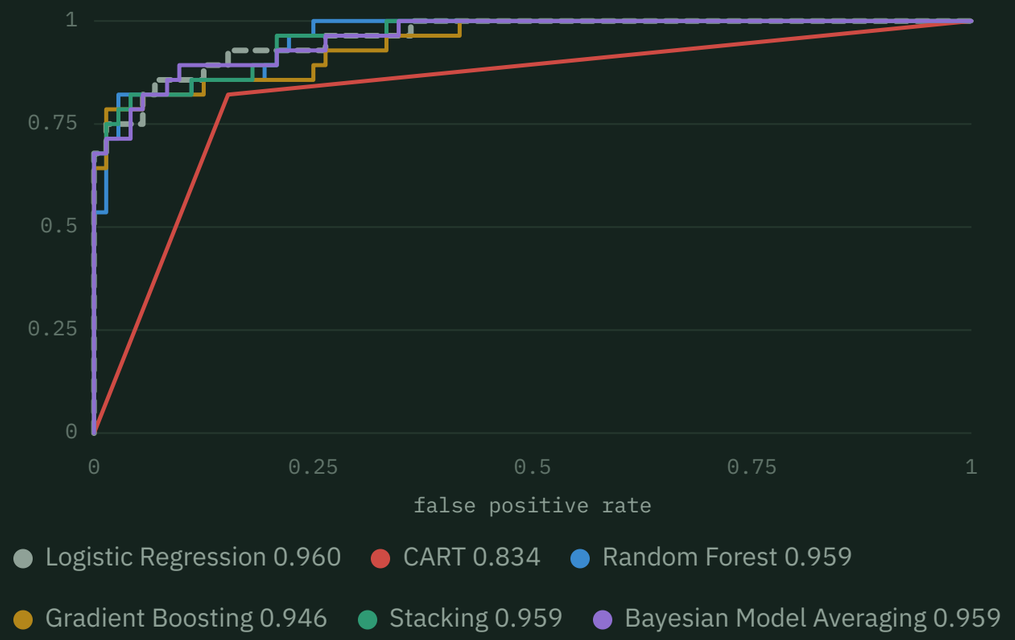
The top of the table is a tie.

REPEATED 5 × 3 CV · CORRECTED RESAMPLED T-TEST

VS LOGISTIC REGRESSION	AUC (15 FOLDS)	Δ	P
Random Forest	0.953 ± 0.018	−0.001	0.90
Gradient Boosting	0.950 ± 0.020	−0.004	0.44
AdaBoost	0.944 ± 0.022	−0.010	0.19
Stacking	0.955 ± 0.016	+0.001	0.78
Soft Voting	0.954 ± 0.018	+0.000	0.95

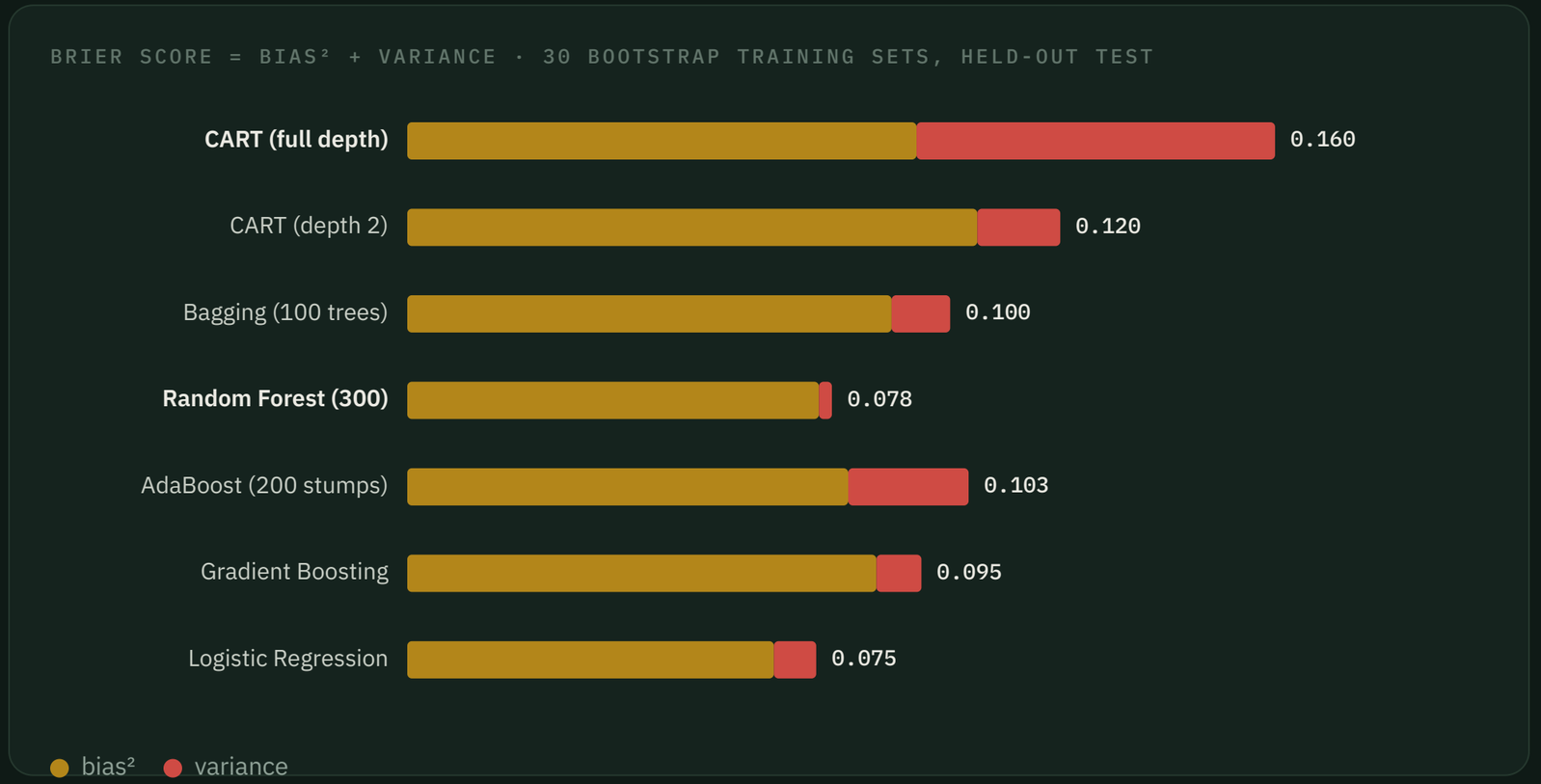
p > 0.05 everywhere: no ensemble is significantly better or worse than logistic regression.

ROC ON THE HELD-OUT TEST SET



RESULT INTERPRETATION · BIAS–VARIANCE

Ensembles kill variance. Logistic had little to begin with.



$$E[(\hat{p} - y)^2] = (\bar{p} - y)^2 + E[(\hat{p} - \bar{p})^2]$$

VARIANCE

full tree **0.066** → random forest **0.002**

READING

Averaging fixes high-variance learners. It cannot fix bias, and boosting attacks bias at the price of some variance.

RESULT INTERPRETATION · FEATURES AND THRESHOLD

Three importance measures, one story.

TOP FEATURES

FEATURE	GB PERMUTATION (ΔAUC)	RF IMPURITY	|LOGISTIC W|
internal marks	0.154	0.299	1.672
attendance %	0.052	0.181	1.812
concern	0.009	0.038	0.519
study hours	0.009	0.017	0.640
CGPA	0.008	0.106	0.517
prev. SGPA	0.007	0.139	0.017
confidence	0.006	0.059	0.752

EARLY-WARNING THRESHOLD · BAYESIAN MODEL AVERAGING

THRESHOLD	RECALL	PRECISION	F1
0.5	82%	85%	0.84
0.4	82%	82%	0.82
0.3	89%	74%	0.81
0.2	93%	60%	0.73

A flag only triggers an **offer of support**, so a missed student costs more than an extra conversation.

RESULT INTERPRETATION

What combining models bought us, and what it didn’t.

FINDINGS

- Averaging trees works: one tree 0.821 → random forest 0.953 CV AUC, because variance falls from 0.066 to 0.002.
- No ensemble beats logistic regression (0.953) by a significant margin: the smallest corrected p-value is 0.19. The best, Bayesian Model Averaging, reaches 0.954.
- So the at-risk signal here is close to additive in the survey answers. Ensembles pay off for high-variance learners, not for an already well-specified model.
- BMA is sure about grades and attendance (inclusion 1.00, 1.00) and drops study habits (0.04).
- Shrinkage matters: with v = 1 boosting over-fits after 8 trees; v = 0.03 gives the lowest test log-loss (0.261).

LIMITS & NEXT STEPS

- The cohort is **synthetic**: its generator builds answers additively from a few latent traits, which suits logistic regression.
- 400 students and 15 features: small data limits how much complex ensembles can learn.
- Real responses with outcome labels may contain interactions where trees and boosting win.
- Prefer the simpler model when scores tie: logistic regression or BMA are easier to explain to a student.

LIVE DEMO

Thank you.
Questions?

Dashboard · localhost:8501/unit-5

docker compose up

python run_all.py · 109 s on a laptop CPU

earlydrop/unit5_ensembles.py